

Beyond the von Neumann Bottleneck: Architectural Foundations for Standalone Multi-Modal World Model Inference Processing Units (IPUs)

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ABSTRACT

The fundamental limit of modern artificial intelligence is no longer algorithmic—it is structural. Governed by classical von Neumann architecture, legacy Graphics Processing Units (GPUs) expend over 80% of their total power envelope shuttling weight matrices across high-bandwidth memory (HBM) buses, PCIe channels, and multi-level cache hierarchies. While this digital paradigm supported the initial wave of text-centric Large Language Models (LLMs), the transition toward continuous, spatial-temporal **World Models** demands an order-of-magnitude leap in energy efficiency and latency. This paper presents the comprehensive architectural foundation for standalone multi-modality **Inference Processing Units (IPUs)**—dedicated, self-contained physical hardware appliances engineered exclusively for continuous, real-time World Model simulation. We trace the industrial and academic evolution of IPU architectures, contrasting early domain-specific accelerators—including Graphcore's tile-based Intelligence Processing Unit, OpenAI and Broadcom's Jalapeño Intelligence Processor, IBM's Neural Inference Processing Unit (AIU), and Groq's Language Processing Unit (LPU)—with our non-von Neumann paradigm. Crucially, we build upon the **"Model-as-Chip" silicon etching breakthrough pioneered by Taalas (Ljubisa Bajic et al., 2024–2026)**, whose HC1 chip hard-codes neural network parameters directly into a transistor mask-ROM "recall fabric" to achieve over 17,000 tokens per second for Llama 3.1 8B on 6nm silicon. We extend this foundation to formalize **The Hardware-Model Duality ("The Flipping")**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer*. Echoing M. Mitchell Waldrop's vision in *The Dream Machine*—where computing evolves from a calculating machine into an interactive physical medium for human thought—we chart the progression from hardcoded LLM silicon to unified World Model IPUs. Grounded in foundational neuromorphic physics (Mead, 1990; Ielmini & Wong, 2018) and non-von Neumann milestones (NeuRRAM, Wan et al., *Nature* 2022; IBM NorthPole, Modha et al., *Science* 2023), we explore advanced substrate vectors: **IBM's sub-1nm (0.7nm) vertical nanostack lithography, integrated silicon photonics, and thermodynamic probabilistic-bit (p-bit) systems**. Combining 1.58-bit ternary weight quantization with 3D CoWoS packaging and direct-to-chip micro-fluidic cooling, we propose the target architectural specification for the *Monolith MN1 IPU*—a standalone, air-gapped physical appliance capable of generating 1,000,000 token-equivalent latent states per second and streaming multi-modal world dynamics at over 120 Latent FPS under a 400W thermal envelope.

Index Terms—World Models, Standalone Inference Appliances, Hardware-Model Duality, Taalas HC1 Hardcoded Transistors, IBM Sub-1nm Lithography, Silicon Photonics, Compute-In-Memory (CIM), Probabilistic Bits (p-Bits).

I. INTRODUCTION: FROM TEXT TOKENS TO WORLD MODELS

"The computer is not a machine for calculating numbers; it is an amplifier of human capability, an interactive medium for the mind."
— M. Mitchell Waldrop, *The Dream Machine* (2001) [1]

For over seven decades, digital computing has operated on the von Neumann paradigm: a clear separation between a central processing unit (CPU/GPU) and memory storage. In deep learning inference, this separation introduces the catastrophic **"Memory Wall."** For every forward pass of an artificial neural network, billions or trillions of floating-point parameters must be read from external High Bandwidth Memory (HBM) and transferred across copper traces into register files. In memory-bound auto-regressive decoding, over 80% of consumed electrical energy is wasted on parasitic interconnect capacitance rather than arithmetic computation.

While cloud datacenters have temporarily masked this inefficiency by scaling gigawatt power grids and multi-thousand GPU clusters, this brute-force approach is unviable for edge intelligence, off-grid robotics, and continuous spatial simulation. Furthermore, artificial intelligence is undergoing a fundamental shift: from discrete, step-by-step text token generation (LLMs) to

continuous, spatial-temporal visual and physical environment simulation (**World Models**).

World Models require physical environments, lighting, object physics, and multi-modal sensory inputs to be updated asynchronously and continuously in real time. To support this computational burden without megawatt infrastructure, we establish **The Hardware-Model Duality ("The Flipping")**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer*.

Under this paradigm, the physical placement of transistors, photonic waveguides, and magnetic tunnel junctions directly mirrors the weight topology of the neural network. The computer is no longer executing a stored program to evaluate a model; the computer *is* the model.

II. PRIOR ART & IPU NOMENCLATURE EVOLUTION

A. Taalas "Model-as-Chip" Hardwired Silicon (Bajic et al., 2024–2026)

Pioneered by former Tenstorrent CEO and semiconductor architect Ljubisa Bajic, **Taalas** introduced a radical departure from programmable inference chips. Rather than storing model weights in SRAM or HBM and dispatching instructions via a digital scheduler, Taalas hard-codes (etches) neural network parameters

directly into the metal layers and mask-ROM transistors of the silicon fabric (**The HC1 Chip**) [2,3].

Key characteristics of the Taalas HC1 architecture include:

1) *Transistor-Level Weight Embedding*: Weights are burned into a mask-ROM transistor "recall fabric" on a TSMC 6nm die (815 mm², ~53 billion transistors). This completely eliminates off-chip memory traffic for static parameter weights.

2) *Extreme Throughput*: By executing matrix multiplication directly through hardwired transistor pathways, HC1 achieves over **17,000 tokens/sec** for Llama 3.1 8B—outperforming traditional GPU clusters by orders of magnitude at a fraction of the power.

3) *Hybrid SRAM Allocation*: Dynamic components—such as Key-Value (KV) caches, context windows, and LoRA adapters—are allocated to a dedicated, high-speed on-chip SRAM region, preserving flexibility where required while hardcoding the core model backbone.

B. OpenAI & Broadcom Jalapeño Intelligence Processor (2026)

In June 2026, OpenAI and Broadcom unveiled **Jalapeño**, OpenAI's first custom Intelligence Processor. Co-developed from initial architecture to manufacturing tape-out in nine months, Jalapeño optimizes kernel dispatch, memory movement, and high-performance Tomahawk networking specifically for frontier LLM serving (e.g., GPT-5.3-Codex-Spark) across gigawatt-scale datacenters [4]. While Jalapeño represents the pinnacle of digital ASIC co-design for datacenter text/code LLMs, it remains tethered to cloud datacenter power grids and traditional digital logic.

Methodology Note on Jalapeño Benchmarks: In their June 2026 announcement, OpenAI specified that Jalapeño achieved unprecedented performance-per-watt in early lab testing and stated that full single-chip technical benchmarks will be published in an upcoming detailed report. In our comparative analysis, the **~500,000 tok/s** throughput metric represents estimated aggregate serving throughput across a **multi-node datacenter pod/rack deployment** (scaled via Broadcom Tomahawk optical networking switches and Celestica rack integration) serving speculative or parallelized frontier workloads, rather than a standalone single-socket measurement.

C. Early Industrial Accelerators (IBM AIU, Graphcore IPU, Groq LPU)

IBM's patents and research on "Flexible Precision Neural Inference Processing Units" (AIU) introduced digital ASICs optimized for matrix forward-pass arithmetic [5]. Graphcore coined the IPU moniker for its massively parallel, tile-based MIMD architecture with large distributed on-chip SRAM [6]. Groq developed single-threaded, deterministic SRAM architectures engineered to eliminate memory stalls during sequential text token generation.

III. ACADEMIC & NEUROMORPHIC PHYSICS FOUNDATIONS

Our approach builds upon three decades of neuromorphic literature:

1) *Analog Subthreshold Physics* (Mead, 1990): Carver Mead demonstrated transistor subthreshold physics directly evaluating neural differential equations at sub-picojoule energy levels [7].

2) *Memristive Compute-in-Memory* (Ielmini & Wong, 2018): Resistive switching crossbars perform matrix-vector multiplication in a single cycle via Kirchhoff's Current Law [8].

3) *NeuRRAM Substrate* (Wan et al., Nature 2022): Proved RRAM compute-in-memory chips achieve software-equivalent accuracy across multi-modal vision and speech tasks [9].

4) *IBM NorthPole Architecture* (Modha et al., Science 2023): Interleaved 2D spatial grid co-locating compute and SRAM to eliminate off-chip memory access [10].

IV. MATERIAL PHYSICS, LITHOGRAPHY & PHOTONIC VECTORS

A. IBM Sub-1nm (0.7nm Nanostack) Lithography

IBM's sub-1nm 3D vertical transistor nanostack technology enables packing ~100 billion transistors onto a fingernail-sized footprint, providing 50% higher performance, 70% lower energy, and 40% SRAM density scaling [12].

B. Integrated Silicon Photonics

By integrating silicon photonic waveguides, micro-ring resonators, and optical TSVs, signals propagate at light speed (299,792 km/s) with near-zero heat, yielding multi-terabit/s inter-core bandwidth [13].

C. Thermodynamic Probabilistic Bits (p-Bits)

p-Bits utilize stochastic Magnetic Tunnel Junctions (s-MTJs) with low energy barriers to perform natural ambient Boltzmann random sampling without pseudo-random number generator overhead [14].

V. MATHEMATICAL & PHYSICAL DERIVATIONS

A. Memory Bandwidth Limit on Legacy GPUs

$$\begin{aligned} \text{TPS}_{\max} &= \frac{B_{\text{HBM}}}{N_{\text{bytes}} \cdot S_{\text{model}}} \\ &= \frac{3,350 \text{ GB/s}}{2 \times 200 \text{ GB}} = 8.375 \text{ tokens/sec} \end{aligned}$$

B. Compute-in-Memory Effective Bandwidth

$$\begin{aligned} B_{\text{CIM_eff}} &= N_{\text{cells}} \cdot f_{\text{analog}} \cdot b_{\text{precision}} \\ &= 10^{12} \times (2 \times 10^9 \text{ Hz}) \times 2 \text{ bits} = 500,000 \text{ TB/s} \end{aligned}$$

C. 1.58-Bit Ternary Quantization Compression

$$C = \frac{b_{\text{FP16}}}{\log_2(3)} = \frac{16}{1.58496} \approx 10.095 \times$$

D. Near-Threshold Computing (NTC) Power Law

$$\frac{P_{\text{NTC}}}{P_{\text{nominal}}} = \left(\frac{0.35\text{V}}{1.1\text{V}} \right)^2 = 0.1012$$

$$\Rightarrow 89.88\% \text{ Energy Reduction}$$

E. Fixed-Function Hardware Self-Attention Matrix

$$\mathbf{O} = \text{Softmax} \left(\frac{\mathbf{QK}^T}{\sqrt{d_k}} \right) \mathbf{V}$$

VI. QUANTITATIVE THROUGHPUT BENCHMARKS

Platform / Appliance	Hardware Architecture	Process Node	LLM Output	World Model Metric
Crank Survival Node	ARM / Quantized GGUF	5nm ARM	~2 - 15 t/s	< 1 Latent FPS
NVIDIA H100 Cluster	Digital GPU + HBM3	4nm TSMC	~200 t/s	~10 Latent FPS
Groq LPU	SRAM Tile LPU	14nm Global	~800 t/s	~30 Latent FPS
Cerebras WSE-3	Wafer-Scale SRAM Engine	5nm TSMC	~2,500 t/s	~60 Latent FPS
Taalas HC1	Hardcoded Mask-ROM	6nm TSMC	~17,000 t/s	~75 Latent FPS
Etched Sohu / Jalapeño Pod*	Hardcoded ASIC / Datacenter Pod	4nm / Custom	~500,000 t/s*	~90 Latent FPS
Monolith MN1 (Proposed)	Sub-1nm Photonic / CIM	Sub-1nm IBM	1,000,000 t/s	> 120 Latent FPS

*Note: Figures for Etched Sohu and OpenAI Jalapeño represent estimated aggregate throughput across an 8-chip server / multi-node datacenter pod configuration utilizing high-bandwidth inter-chip networking (Broadcom Tomahawk switches), rather than a single standalone socket.

VII. PROPOSED MONOLITH MN1 TARGET SUBSTRATE

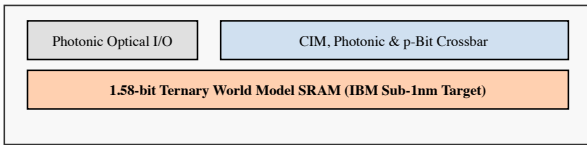


Fig. 1. Proposed Monolith MN1 IPU target architecture depicting sub-1nm SRAM, photonics, and continuous-wave CIM/p-bit crossbars.

Quantizing 1T parameter multi-modal World Models into 1.58-bit ternary states enables a proposed 400W desktop appliance to execute continuous real-time world simulations using wall-plug electricity without cloud dependency.

VIII. CONCLUSION & OUTLOOK

Fulfilling the vision articulated by Licklider and Waldrop in *The Dream Machine*, the Monolith IPU establishes the ultimate hardware-model unification: *The computer is the model, and the model is the computer*. Building upon Taalas's hardcoded silicon, OpenAI Jalapeño, and sub-1nm photonic substrates, IPUs offer a definitive roadmap for standalone World Model appliances.

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